

Sensor Fusion and Data Association Problems

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Date: 2026

PART A: CONCEPTUAL QUESTIONS (Simple to Intermediate)

A1. In precision-weighted fusion, why is it natural to work with precisions $\lambda_i = 1/\sigma_i^2$ rather than variances σ_i^2 ? What happens to the fused estimate when one sensor has $\sigma_i \rightarrow \infty$?

A2. What is *data association*, and why is it needed in multi-target tracking? Give an example scenario where ignoring data association leads to incorrect fusion.

A3. List some benefits and disadvantages of *classical gating* (Mahalanobis distance with a χ^2 threshold).

A4. Define *aleatoric uncertainty* and *epistemic uncertainty* in the context of multi-sensor fusion. For each type, give one example of a sensor condition that primarily increases that type of uncertainty, and explain how each type affects the fusion output.

PART B: ANALYTICAL PROBLEMS — TRADITIONAL FUSION

B1. Two-Sensor Precision-Weighted Fusion

Learning objective: Apply the MLE fusion formula and verify its properties by hand.

Problem. Two independent sensors measure the same scalar state z :

$$y_1 = z + v_1, \quad v_1 \sim \mathcal{N}(0, \sigma_1^2), \quad y_2 = z + v_2, \quad v_2 \sim \mathcal{N}(0, \sigma_2^2).$$

Given $\sigma_1 = 2$, $\sigma_2 = 1$, observed values $y_1 = 4.8$, $y_2 = 5.3$, and true value $z = 5$.

- (a) Compute the precisions λ_1 , λ_2 , and the fused precision λ_f .
- (b) Compute the fusion weights w_1 , w_2 and verify $w_1 + w_2 = 1$.
- (c) Compute the fused estimate $\hat{z}$ and the fused variance σ_f^2 .
- (d) Verify that $\sigma_f < \min(\sigma_1, \sigma_2)$.
- (e) Compute the absolute error for each sensor and for the fused estimate. Which is smallest?
- (f) Compute the information gain $\Delta I = \frac{1}{2} \log(\sigma_1^2/\sigma_f^2)$ in nats. Interpret.

B2. Kalman Filter Multi-Sensor Update

Learning objective: Walk through a single KF measurement update step, computing all intermediate quantities.

Problem. A Kalman filter tracks a 2-D state $\mathbf{z} = [p, v]^T$ (position and velocity). After the predict step:

$$\boldsymbol{\mu}_{t|t-1} = \begin{pmatrix} 10 \\ 2 \end{pmatrix}, \quad \boldsymbol{\Sigma}_{t|t-1} = \begin{pmatrix} 5 & 1 \\ 1 & 2 \end{pmatrix}.$$

A GPS sensor measures position only: $\mathbf{H} = [1 \ 0]$, $R = 4$. The observed measurement is $y = 12$.

- Compute the innovation $\tilde{y} = y - \mathbf{H}\boldsymbol{\mu}_{t|t-1}$.
- Compute the innovation covariance $S = \mathbf{H}\boldsymbol{\Sigma}_{t|t-1}\mathbf{H}^T + R$.
- Compute the Kalman gain $\mathbf{K} = \boldsymbol{\Sigma}_{t|t-1}\mathbf{H}^T S^{-1}$.
- Compute the updated mean $\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}\tilde{y}$.
- Compute the updated covariance $\boldsymbol{\Sigma}_{t|t} = (\mathbf{I} - \mathbf{K}\mathbf{H})\boldsymbol{\Sigma}_{t|t-1}$.
- Verify that $\det(\boldsymbol{\Sigma}_{t|t}) < \det(\boldsymbol{\Sigma}_{t|t-1})$. What does this mean physically?
- If a second sensor (IMU) measuring velocity arrives with $\mathbf{H}_2 = [0 \ 1]$, $R_2 = 1$, $y_2 = 2.5$, apply a sequential update starting from $(\boldsymbol{\mu}_{t|t}, \boldsymbol{\Sigma}_{t|t})$. Compute the new posterior.

B3. Gating with Mahalanobis Distance (Continuation of the previous problem)

Learning objective: Compute Mahalanobis distances and determine which measurements pass an ellipsoidal gate.

Problem. A Kalman filter predicts measurement $\hat{\mathbf{y}} = (10, 5)^T$ with innovation covariance

$$\mathbf{S} = \begin{pmatrix} 4.25 & 3.0 \\ 3.0 & 5.25 \end{pmatrix}, \quad \mathbf{S}^{-1} = \begin{pmatrix} 0.394 & -0.225 \\ -0.225 & 0.319 \end{pmatrix}.$$

The gating threshold is $P_G = 0.99$ with $n_y = 2$ measurement dimensions, giving $\gamma = \chi_2^2(0.99) = 9.21$.

Four candidate measurements are received:

Point	Measurement $\mathbf{y}$
P_1	(13, 5)
P_2	(10, 8)
P_3	(12, 7)
P_4	(15, 1)

- For each point, compute the innovation $\tilde{\mathbf{y}} = \mathbf{y} - \hat{\mathbf{y}}$.
- For each point, compute the squared Mahalanobis distance $d_M^2 = \tilde{\mathbf{y}}^T \mathbf{S}^{-1} \tilde{\mathbf{y}}$.
- For each point, decide: inside gate ($d_M^2 \leq 9.21$) or outside?
- Point P_4 has Euclidean distance $\|\tilde{\mathbf{y}}\| = 6.4$, while P_1 has $\|\tilde{\mathbf{y}}\| = 3.0$. Yet P_1 passes and P_4 does not. Explain why the Mahalanobis distance gives a different answer than Euclidean distance.
- If we lower the gating probability to $P_G = 0.95$ (giving $\gamma = 5.99$), which points would now be rejected? What is the trade-off?

B4. JPDA: Joint Association Probabilities

Learning objective: Enumerate joint association events and compute marginal association weights for a small tracking scenario.

Problem. Two targets T_1, T_2 are being tracked. At the current scan, two measurements $\mathbf{y}^{(1)}, \mathbf{y}^{(2)}$ fall inside both gates. Assume detection probability $P_D = 0.9$.

The individual measurement likelihoods are:

	$\mathbf{y}^{(1)}$	$\mathbf{y}^{(2)}$
T_1	$\mathcal{L}_{11} = 0.50$	$\mathcal{L}_{12} = 0.05$
T_2	$\mathcal{L}_{21} = 0.08$	$\mathcal{L}_{22} = 0.40$

(These are the values of $\mathcal{N}(\mathbf{y}^{(j)}; \hat{\mathbf{y}}_i, \mathbf{S}_i)$ for track i and measurement j .)

For simplicity, consider only two joint events:

- $\Theta_1: T_1 \leftarrow \mathbf{y}^{(1)}, T_2 \leftarrow \mathbf{y}^{(2)}$ (“straight” assignment)
- $\Theta_2: T_1 \leftarrow \mathbf{y}^{(2)}, T_2 \leftarrow \mathbf{y}^{(1)}$ (“swapped” assignment)

- Compute the unnormalised score for each joint event: $w(\Theta_k) = P_D^2 \prod_i \mathcal{L}_{i,j(i)}$.
- Normalise to get $P(\Theta_1)$ and $P(\Theta_2)$.
- Compute the marginal association probability $\beta_{1,1} = P(\text{Track 1 gets } \mathbf{y}^{(1)})$.
- Compute $\beta_{2,1} = P(\text{Track 2 gets } \mathbf{y}^{(1)})$.
- Verify $\beta_{1,1} + \beta_{2,1} \leq 1$. Compare this with what PDAF would give (where each track ignores the other). Why does JPDA prevent track coalescence?
- If $\mathcal{L}_{11}$ and $\mathcal{L}_{21}$ were both 0.30 (ambiguous), what would happen to the marginal probabilities? Compute.

PART C: ANALYTICAL PROBLEMS — DEEP FUSION

C1. Cosine Similarity vs. Euclidean Distance for Embeddings

Learning objective: Understand why cosine similarity is preferred over Euclidean distance in high-dimensional embedding spaces.

Problem. A CNN produces 2-D embeddings for three objects:

- Car in bright light: $\mathbf{a} = (4, 3)$
- Same car in shadow: $\mathbf{b} = (1.2, 0.9)$
- Truck: $\mathbf{c} = (-2, 3.5)$

- Compute $\|\mathbf{a} - \mathbf{b}\|_2$ and $\|\mathbf{a} - \mathbf{c}\|_2$. Which pair appears more similar by Euclidean distance?
- Compute $\text{Sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$ and $\text{Sim}(\mathbf{a}, \mathbf{c})$.
- Which metric correctly identifies that $\mathbf{a}$ and $\mathbf{b}$ are the same object? Why?

- (d) In 256-dimensional embedding space, Euclidean distances tend to concentrate. Explain briefly why cosine similarity is more robust to this “curse of dimensionality.”

C2. Heteroscedastic PoE Fusion under Sensor Corruption

Learning objective: Apply Product-of-Experts fusion with learned variances, and show how corruption is automatically handled.

Problem. Two modality networks each output a Gaussian predictive distribution $\mathcal{N}(\mu_m, \sigma_m^2)$ for a target position.

Normal conditions:

Modality	μ_m	σ_m^2
IMU (A)	4.8	0.5
Camera (B)	5.1	0.8

Corrupted conditions (camera blinded by glare):

Modality	μ_m	σ_m^2
IMU (A)	4.8	0.5
Camera (B)	7.2	12.0

Gaussian PoE fusion: $\sigma_f^{-2} = \sigma_A^{-2} + \sigma_B^{-2}$, $\mu_f = \sigma_f^2(\sigma_A^{-2}\mu_A + \sigma_B^{-2}\mu_B)$.

- Compute μ_f and σ_f^2 under normal conditions.
- Compute the weight (fraction of total precision) contributed by each modality under normal conditions.
- Compute μ_f and σ_f^2 under corrupted conditions.
- Compute the weight contributed by each modality under corrupted conditions.
- How much did the fused estimate shift toward the corrupted camera reading? Explain why the shift is small.
- What would happen if both modalities always reported $\sigma_m^2 = 1$ (homoscedastic, ignoring corruption)? Compute μ_f in the corrupted case and compare.

PART D: INTERPRETATION AND PRACTICAL QUESTIONS

D1. Interpreting Mixture-of-Experts Gate Weights

A gated fusion network was trained with modality dropout ($p_{\text{drop}} = 0.3$) on two sensor modalities (A and B). The following gate weights were observed at test time:

Condition	g_A	g_B	Test RMSE
Both normal	0.58	0.42	0.32
B corrupted ($\sigma = 5$)	0.91	0.09	0.61
B corrupted ($\sigma = 20$)	0.95	0.05	0.65
A-only baseline	1.00	0.00	0.58

- (a) Why does g_A increase as modality B becomes more corrupted?
- (b) When B is severely corrupted ($\sigma = 20$), the RMSE is 0.65, which is *worse* than the A-only baseline (0.58). Why?
- (c) If the network had been trained *without* modality dropout, what would you expect the gate weights and RMSE to look like under corruption?
- (d) How does this gating mechanism relate to classical precision-weighted fusion?

D2. MC Dropout and Epistemic Uncertainty for Fusion

A fusion network uses MC dropout ($T = 50$ forward passes) at test time. For in-distribution test inputs the mean predictive variance is 0.12. For out-of-distribution (OOD) inputs (sensor shifted to a new domain), the mean predictive variance is 2.8.

- (a) Is the variance from MC dropout primarily measuring aleatoric or epistemic uncertainty? Justify.
- (b) Why is the OOD variance much larger?
- (c) A deep ensemble ($M = 5$ independently trained networks) gives in-distribution variance 0.10 and OOD variance 4.1. Why might the ensemble detect OOD inputs more reliably than MC dropout?
- (d) In a safety-critical fusion system, how would you use these variance estimates to decide whether to trust the deep fusion output or fall back to a classical Kalman filter?

D3. Missing Modality Robustness

Consider three fusion architectures: (i) Early fusion (concatenate raw inputs), (ii) Late fusion (combine per-modality predictions), (iii) Gated MoE (learned reliability weights, trained with modality dropout).

- (a) If modality B is completely missing at test time (all zeros), which architecture(s) will fail catastrophically? Why?
- (b) In the classical information filter, a missing modality is handled by simply omitting its precision contribution. Write the formula and explain why this is natural.
- (c) How does modality dropout during training achieve a similar effect in the deep setting?

PART E: SYNTHESIS PROBLEMS

E1. Correlated Sensors and Covariance Intersection

Learning objective: Show that ignoring sensor correlation leads to overconfidence, and apply CI as a remedy.

Problem. Two sensors measure the same scalar z with $\sigma_1 = \sigma_2 = 2$, but their noise is correlated: $\text{Cov}(v_1, v_2) = \Sigma_{12} = 1$ (from a shared vibration source).

- (a) If the sensors were independent, what would the fused variance be?
- (b) Compute the correct fused variance accounting for correlation: $\sigma_f^2 = (\sigma_1^2 \sigma_2^2 - \Sigma_{12}^2) / (\sigma_1^2 + \sigma_2^2 - 2\Sigma_{12})$.
- (c) By how much does the naive formula underestimate uncertainty?
- (d) If Σ_{12} is unknown but positive, Covariance Intersection uses: $\sigma_f^{-2} = \omega \sigma_1^{-2} + (1 - \omega) \sigma_2^{-2}$. For $\omega = 0.5$, compute σ_f^2 . Is it larger or smaller than the correct answer from (b)?
- (e) Explain why a conservative (larger) uncertainty is preferable to an overconfident one in a safety-critical tracking system.

E2. End-to-End Comparison: Stress Testing Fusion Methods

Learning objective: Interpret a corruption stress test and draw engineering conclusions about robustness in multimodal fusion.

Problem. The following RMSE values were measured on a regression task as noise is added to modality B:

σ_{corrupt}	0	1	2	5	10	20
Early Fusion (concat)	0.35	0.48	0.72	2.10	4.80	9.50
Gated MoE (mod. dropout)	0.32	0.34	0.38	0.52	0.57	0.58
Heteroscedastic PoE	0.30	0.33	0.35	0.45	0.54	0.58
A-only baseline	0.58	0.58	0.58	0.58	0.58	0.58

- (a) At $\sigma_{\text{corrupt}} = 0$, all multimodal methods beat the A-only baseline. Why?
- (b) At which corruption level does early fusion become *worse* than using modality A alone?
- (c) The Gated MoE approaches the A-only baseline as corruption grows. Why is this the desired behavior?
- (d) The heteroscedastic PoE is the best at high corruption. Connect this to the precision-weighting principle from Part B1.
- (e) If you were designing a safety-critical autonomous driving system, which method would you deploy and why? What additional safeguard would you add?

PART F: Problems related to the associated notebook

1. F1. Sequential Kalman fusion: IMU every step, GPS every 5 steps

In the notebook, a 1D constant-velocity Kalman filter tracks state

$$\mathbf{x}_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}, \quad \mathbf{x}_k = \mathbf{F} \mathbf{x}_{k-1} + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}),$$

with

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \quad \mathbf{Q} = 0.5 \begin{bmatrix} \Delta t^3/3 & \Delta t^2/2 \\ \Delta t^2/2 & \Delta t \end{bmatrix}, \quad \Delta t = 0.1.$$

Two sensors are fused *sequentially*:

$$\text{IMU: } z_k^{\text{imu}} = \mathbf{H}_{\text{imu}} \mathbf{x}_k + v_k^{\text{imu}}, \quad \mathbf{H}_{\text{imu}} = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad R_{\text{imu}} = 0.5^2,$$

$$\text{GPS: } z_k^{\text{gps}} = \mathbf{H}_{\text{gps}} \mathbf{x}_k + v_k^{\text{gps}}, \quad \mathbf{H}_{\text{gps}} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad R_{\text{gps}} = 2.0^2,$$

where GPS arrives only when $k \bmod 5 = 0$. The notebook reports (last time step):

$$|p_T - \hat{p}_T| = 0.434 \text{ m}, \quad |v_T - \hat{v}_T| = 0.225 \text{ m/s}, \quad \text{tr}(\mathbf{P}_T) = 0.3681.$$

- (a) Conceptually, why would an *IMU-only* filter tend to drift in position, even if velocity is measured every step?
- (b) The covariance trace $\text{tr}(\mathbf{P}_k)$ typically shows a *sawtooth* pattern (decreasing at GPS steps). Explain why.
- (c) Show that if the two measurements at time k are independent, then doing an IMU update and then a GPS update is equivalent (in information content) to a *single joint update* using a stacked measurement.

2. F2. Product of Experts fusion and missing modality

The notebook fuses three scalar modality estimates using a Product-of-Experts (PoE) rule:

$$p(x \mid \text{all}) \propto \prod_{m=1}^M \mathcal{N}(x; \mu_m, \sigma_m^2).$$

It uses the following *means and variances*:

$$(\mu, \sigma^2)_{\text{IMU}} = (4.8, 0.50), \quad (\mu, \sigma^2)_{\text{GPS}} = (5.3, 1.50), \quad (\mu, \sigma^2)_{\text{Cam}} = (5.1, 0.80).$$

- (a) Compute the fused mean μ_f and variance σ_f^2 .
- (b) GPS drops out (missing modality). Recompute (μ_f, σ_f^2) and explain qualitatively why the uncertainty increases and why the mean shifts.

3. F3. Crossing targets: why GNN, PDAF, JPDA have similar RMSE here

The notebook simulates two crossing targets with clutter, then compares: *GNN* (*hard assignment*), *PDAF* (*soft per-track*), *JPDA* (*soft joint*). It reports essentially identical RMSE:

$$\text{Avg RMSE} \approx 0.69 \text{ for GNN, PDAF, JPDA.}$$

Yet the qualitative table says (in general) GNN can be poor at crossing, PDAF can coalesce, JPDA is good.

- (a) Explain why all three methods can achieve nearly identical RMSE in this particular run.
- (b) Conceptually, what failure mode is the table warning about for (i) GNN and (ii) PDAF at crossing?
- (c) Predict what would happen if clutter intensity λ_c increases or detection probability P_D decreases. Which method would you expect to degrade first, and why?